CS 6515: Graduate Algorithms

Lecture 13: Projected and Smooth Gradient Descent

Instructors: Sahil Singla and Gerandy Brito

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Recap and plan. For convex f, bounded gradients, and $\|x_0 - x^*\|_2 \le D$, Lecture 12 proved

$$f(\bar{x}_T) - f(x^*) \le \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}.$$

With $\eta = D/(G\sqrt{T})$, this is $DG/\sqrt{T}$. Today: keep iterates feasible using projection, then exploit upper curvature bounds to obtain faster convergence for a fixed objective.

1 Constrained optimization: project the step back

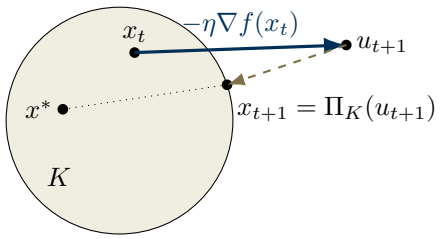
Let $K \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. The Euclidean projection of u onto K is

$$\Pi_K(u) = \arg\min_{z \in K} \|z - u\|_2^2.$$

The minimum exists, and its point is unique by strict convexity of squared distance on a convex set.

Projected gradient descent. Start at $x_0 \in K$, and repeat

$$u_{t+1} = x_t - \eta \nabla f(x_t), \qquad \boxed{x_{t+1} = \Pi_K(u_{t+1}).}$$



The raw gradient step can leave K; projection restores feasibility.

Example. For a box $K = \prod_i [a_i, b_i]$, project by clipping each coordinate:

$$(\Pi_K(u))_i = \min\{b_i, \max\{a_i, u_i\}\}.$$

Projection may itself be a nontrivial optimization problem. Its computational cost must be included in the running time.

1.1 Why projection preserves the distance argument

Theorem (projection cannot increase distance to a feasible point). If $p = \Pi_K(u)$ and $z \in K$, then

$$\boxed{\|p - z\|_2^2 \le \|u - z\|_2^2 - \|u - p\|_2^2 \le \|u - z\|_2^2.}$$

Proof. The segment $p + \theta(z - p)$ stays in K. Because p minimizes distance to u, the right derivative at $\theta = 0$ of $\|u - p - \theta(z - p)\|_2^2$ is nonnegative. Hence

$$\langle u - p, z - p \rangle \le 0.$$

Expand

$$\|u - z\|_2^2 = \|u - p\|_2^2 + \|p - z\|_2^2 + 2\langle u - p, p - z \rangle \ge \|u - p\|_2^2 + \|p - z\|_2^2.$$

Plug it into the existing proof. Take $z = x^* \in K$. Then

$$\|x_{t+1} - x^*\|_2^2 \le \|x_t - \eta \nabla f(x_t) - x^*\|_2^2.$$

Everything after this line is exactly the expansion and telescoping from Lecture 12. Therefore, with $\|\nabla f(x_t)\|_2 \le G$,

$$f(\bar{x}_T) - f(x^*) \le \frac{D^2}{2\eta T} + \frac{\eta G^2}{2}, \qquad \bar{x}_T \in K.$$

The same proof permits bounded subgradients.

2 Smoothness: an upper bound on curvature

For the faster bounds below, we return to **unconstrained** minimization over $\mathbb{R}^d$, with an attained minimum x^*.

Definition (smoothness). A differentiable $f : \mathbb{R}^d \to \mathbb{R}$ is β-smooth if

$$\|\nabla f(y) - \nabla f(x)\|_2 \le \beta \|y - x\|_2 \qquad \text{for all } x, y.$$

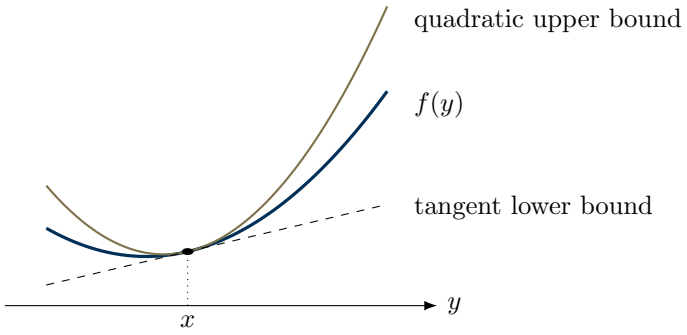
We use a positive upper bound $\beta > 0$.

Theorem (quadratic upper bound, or descent lemma). For a β-smooth function,

$$\boxed{f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2}\|y - x\|_2^2.}$$

Proof. Put $h = y - x$. The fundamental theorem of calculus and Cauchy–Schwarz give

$$\begin{aligned} f(y) - f(x) - \langle \nabla f(x), h \rangle &= \int_0^1 \langle \nabla f(x + sh) - \nabla f(x), h \rangle \, ds \\ &\le \int_0^1 \beta s \|h\|_2^2 \, ds = \frac{\beta}{2}\|h\|_2^2. \end{aligned}$$



For convex twice continuously differentiable functions, smoothness is equivalent to $0 \preceq \nabla^2 f(x) \preceq \beta I$. Smoothness bounds changes in the gradient, *not* its magnitude: $f(x) = x^2/2$ is 1-smooth on $\mathbb{R}$, but has unbounded gradient.

3 A safe step size and guaranteed descent

Choose $\eta = 1/\beta$:

$$x_{t+1} = x_t - \frac{1}{\beta}\nabla f(x_t).$$

Substituting this update into the quadratic upper bound gives

$$\begin{aligned} f(x_{t+1}) &\le f(x_t) - \frac{1}{\beta}\|\nabla f(x_t)\|_2^2 + \frac{\beta}{2}\frac{1}{\beta^2}\|\nabla f(x_t)\|_2^2 \\ &= f(x_t) - \frac{1}{2\beta}\|\nabla f(x_t)\|_2^2. \end{aligned}$$

Thus

$$\boxed{f(x_{t+1}) \le f(x_t) - \frac{1}{2\beta}\|\nabla f(x_t)\|_2^2.} \tag{1}$$

In particular, function values never increase.

Why the general-convex analysis was weaker. With only a gradient bound G, a fixed step need not decrease the function at every iteration. Smoothness controls the error in the linear approximation, allowing a constant safe step size and a last-iterate guarantee.

4 Smooth convex functions: a faster last-iterate bound

Theorem (smooth gradient descent). Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is convex and β-smooth, attains its minimum at x^*, and $\|x_0 - x^*\|_2 \le D$. After $T \ge 1$ steps of size $1/\beta$,

$$\boxed{f(x_T) - f(x^*) \le \frac{\beta D^2}{2T}.}$$

Thus $\mathcal{O}(\beta D^2/\varepsilon)$ iterations suffice for additive error ε.

Proof plan. Use the same squared-distance potential as before. This time the negative gradient-squared term from actual descent cancels the positive one in the distance expansion.

Set $g_t = \nabla f(x_t)$ and $R_t = \|x_t - x^*\|_2^2$. Expanding the update gives the exact identity

$$\frac{\beta}{2}(R_t - R_{t+1}) = \langle g_t, x_t - x^* \rangle - \frac{1}{2\beta}\|g_t\|_2^2.$$

By convexity and (1),

$$\begin{aligned} f(x_{t+1}) - f(x^*) &\le f(x_t) - f(x^*) - \frac{1}{2\beta}\|g_t\|_2^2 \\ &\le \langle g_t, x_t - x^* \rangle - \frac{1}{2\beta}\|g_t\|_2^2 \\ &= \frac{\beta}{2}(R_t - R_{t+1}). \end{aligned}$$

Sum for $t = 0, \dots, T-1$:

$$\sum_{t=0}^{T-1} \big(f(x_{t+1}) - f(x^*)\big) \le \frac{\beta}{2}(R_0 - R_T) \le \frac{\beta D^2}{2}.$$

Function values decrease, so each term on the left is at least $f(x_T) - f(x^*)$. Hence

$$T\big(f(x_T) - f(x^*)\big) \le \frac{\beta D^2}{2},$$

which proves the theorem.

Two improvements over the basic bound. The error is $\mathcal{O}(1/T)$, not $\mathcal{O}(1/\sqrt{T})$, and the guarantee holds for the last iterate x_T, not just an average. No extra $\log T$ factor is needed.

Assumptions	Output and error after T steps	Step size
Convex, gradients bounded by G	Average; $DG/\sqrt{T}$	$D/(G\sqrt{T})$
Convex and β-smooth	Last iterate; $\beta D^2/(2T)$	$1/\beta$

Do not overinterpret the conclusions.

- Convexity alone does not specify an efficient oracle or cheap projection. Running time includes those costs.
- For a nonconvex function, a zero gradient need not be a local minimum. Our global-optimality proofs used convexity essentially.
- A fixed-objective additive-error guarantee is not the same as an approximation ratio for a discrete optimization problem.